

MINIMUM DEGREE EIGHT AND A ROOTED-WEB REDUCTION IN K_7^- -MINOR-FREE CONTRACTION-CRITICAL GRAPHS

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ABSTRACT. Let G be a minor-minimal non-six-colourable K_7^- -minor-free graph, with n_i vertices of degree i , and put $\tau = \sum_{i \geq 10} (i-9)n_i$. We prove

$$n_7 = 0, \quad \delta(G) \geq 8, \quad |E(G)| \geq 4|V(G)|, \quad n_8 \geq 25 + \tau,$$

and show that G has no K_5 subgraph. Moreover, G has an independent four-set U of degree-eight vertices such that $G - U$ is three-connected, nonplanar, and six-chromatic. For $r \in U$ and a six-colouring of $G - r$, choose four colours occurring exactly once on $N_G(r)$. Either their representatives root a K_4 minor in $G - U$, or $G - U$ is a spanning subgraph of a web rooted at them with a seven-vertex separator $U \dot{\cup} T$. In the web case, the three-separations form a lattice after two vertices are fixed on opposite sides, one of two alternating bichromatic components meets $U - \{r\}$, and $G[T] \neq K_3$. Deleting a selected edge yields either five controlled paths or a smaller component behind another seven-vertex separator, with a boundary colour class preserved in the latter case. In either edge-deletion outcome, there is also a six-ended fan whose residual paths, after removal of its centre, include at least two pairs with no edge between them. We also prove that a six-connected n -vertex graph with a degree- d vertex whose neighbourhood contains K_4 and at least $4n + d - 13$ edges contains a K_7^- minor. The proofs are computation-free apart from the cited determination of $R(4, 5)$.

1. INTRODUCTION

Hadwiger's conjecture states that every graph of chromatic number t contains a K_t minor [4]. It is known for $t \leq 6$, with the last case due to Robertson, Seymour and Thomas [11], and is open for every $t \geq 7$. Write K_t^- for the graph obtained from K_t by deleting one edge, and write $K_7^\vee$ for the graph obtained from K_7 by deleting two incident edges. Every K_7^- -minor-free graph is seven-colourable [1, Theorem 1]; Albar's proof uses Jakobsen's extremal theorem. Norin and Totschnig proved that every $K_7^\vee$ -minor-free graph is six-colourable [10, Theorem 4]. They asked whether the same conclusion holds under the weaker exclusion of K_7^- [10, Conjecture 21].

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Neighbourhoods of low-degree vertices in contraction-critical graphs have a long history. Mayer obtained a finite catalogue of candidate neighbourhood graphs for degree-seven vertices in six-contraction-critical graphs [8]. In the minor-minimal $K_7^\vee$ -minor-free setting, Norin and Totschnig proved that every degree-seven vertex belongs to a K_5 contained in its closed neighbourhood [10, Claim 4.4]. Here a rooted-minor argument first determines the entire neighbourhood. A second rooted argument, using their $K_{4,2}^*$ bound, then shows that no degree-seven vertex can occur at all.

Rolek and Song proved the next theorem for $r = 6$ [12, Lemma 1.9]. We prove it for every $r \geq 2$.

Theorem 1.1. *For every $r \geq 2$, every $(r + 1)$ -connected graph containing two distinct K_r subgraphs contains a K_{r+2}^- minor.*

The connectivity hypothesis is sharp: K_{r+1} is r -connected and contains two distinct K_r subgraphs, but has no K_{r+2}^- minor.

The rooted closure below is the main local tool. A graph is *minor-minimal non-six-colourable* if it is not six-colourable and every proper minor is six-colourable.

Theorem 1.2. *Let G be a six-connected graph of order n , let $v \in V(G)$ have degree $d \geq 5$, and suppose that $G[N(v)]$ contains a K_4 subgraph. If*

$$|E(G)| \geq 4n + d - 13,$$

then $K_7^- \preccurlyeq G$.

The closure gives the following consequences for a critical graph.

Theorem 1.3. *Let G be a minor-minimal non-six-colourable K_7^- -minor-free graph. For each i , let n_i denote the number of vertices of degree i , and put*

$$\tau = \sum_{i \geq 10} (i - 9)n_i.$$

Then G is seven-connected and

$$n_7 = 0, \quad \delta(G) \geq 8, \quad |E(G)| \geq 4|V(G)|.$$

Moreover, G contains no K_5 subgraph and

$$n_8 \geq 25 + \tau.$$

Every degree-eight vertex of G has a K_4 -free neighbourhood.

Four independent degree-eight vertices yield a simultaneous reduction. A minor model is *rooted* at nominated vertices when its branch sets contain them one-to-one.

Theorem 1.4. *Let G be as in Theorem 1.3. There is an independent four-set U of degree-eight vertices. Put $H = G - U$. Then H is three-connected, nonplanar, and six-chromatic, and $\kappa(G) \in \{7, 8\}$.*

Fix $r \in U$, a six-colouring φ of $G - r$, and four colours which occur exactly once on $N_G(r)$, with representatives x_1, x_2, x_3, x_4 . Exactly one of the following holds.

- (1) The graph H has a K_4 minor rooted at x_1, x_2, x_3, x_4 . Together with the singleton branch set $\{r\}$, this is a K_5 minor in $G - (U - \{r\})$ with branch sets containing r, x_1, x_2, x_3, x_4 , respectively.
- (2) The graph H is a spanning subgraph of a web rooted at x_1, x_2, x_3, x_4 . A facial three-set T of the containing web has a nonempty connected set C behind it such that

$$(1.1) \quad N_H(C) = T, \quad N_G(C) = U \dot{\cup} T.$$

Here T is not asserted to be a triangle in G ; in fact it is not.

If $\kappa(G) = 8$, only the first outcome occurs. If $\kappa(G) = 7$, every proper three-separation of H lifts to an order-seven separation whose separator consists of U together with its three-vertex separator in H . Oriented between two fixed vertices lying on opposite open sides of every separation, these separations form a meet/join lattice. If three such separations have all eight sign regions nonempty, each member of U has exactly one neighbour in each region; four separations cannot have all sixteen sign regions nonempty. Moreover, in the web outcome an alternating bichromatic component passes through a member of $U - \{r\}$.

Albar proved the analogous lower bound of twenty-five degree-eight vertices for a hypothetical minor-minimal non-seven-colourable K_7^- -minor-free graph [1, Lemma 6]. Theorem 1.3 concerns criticality one level lower: the rooted closure eliminates degree seven and gives the defect-sensitive bound $n_8 \geq 25 + \tau$.

A *minor model* of a graph H in G is a family $(B_x : x \in V(H))$ of pairwise disjoint connected vertex sets such that B_x and B_y are adjacent whenever $xy \in E(H)$. We use $H \preceq G$ to denote that H is a minor of G . All graphs are finite and simple.

2. TWO LINKED CLIQUES

Lemma 2.1. *Let $r \geq 2$. If an $(r+1)$ -connected graph H contains two vertex-disjoint K_r subgraphs, then $K_{r+2}^- \preceq H$. The branch sets can be chosen to meet the union of the two cliques.*

Proof. Write the two cliques as

$$A = \{a_1, \dots, a_r\}, \quad B = \{b_1, \dots, b_r\}.$$

By Menger's theorem there are r pairwise vertex-disjoint A to B paths whose open interiors avoid $A \cup B$. Relabel them as paths P_i from a_i to b_i , and put $U = \bigcup_i V(P_i)$.

Call a path Q a *usable cross-path* if its ends lie on distinct paths P_i, P_j , its open interior avoids U , and its ends are not both A -ends or both B -ends. Suppose first that Q exists. Split each of P_i and P_j across an edge into a

nonempty A -prefix and a nonempty B -suffix, choosing the split so that the ends of Q belong to pieces of opposite kinds. Absorb the open interior of Q into the piece containing one end. The four resulting pieces are connected and disjoint. The two A -pieces are adjacent through A , the two B -pieces are adjacent through B , the two pieces from each original path are adjacent across its splitting edge, and Q supplies one of the two remaining cross-adjacencies. Thus at most one adjacency among the four pieces is missing. Each unsplit path is adjacent to every split piece through its two clique ends, and the unsplit paths are pairwise adjacent. These $r + 2$ sets form a K_{r+2}^- model.

Suppose now that no usable cross-path exists. Let C be a component of $H - U$. If C has neighbours on two distinct paths, a path through C between two such attachments is unusable only when both attachments are A -ends or both are B -ends. Comparing all pairs of attachments shows that all neighbours of C are A -ends or all are B -ends. Hence $|N_H(C)| \leq r$, contrary to $(r + 1)$ -connectivity. Every component of $H - U$ therefore attaches to only one P_i .

For each i , let W_i be the union of the open interior of P_i and all components of $H - U$ that attach to P_i . No vertex of W_i has a neighbour on P_j for $j \neq i$. Indeed, an edge from the open interior of P_i to P_j would be a usable cross-path, while such an edge from a component of $H - U$ would make that component attach to two distinct linkage paths, contrary to the preceding paragraph. Hence

$$N_H(W_i) \subseteq \{a_i, b_i\}.$$

Since $r \geq 2$, a nonempty W_i would give a vertex cut of order at most two. Thus every W_i is empty. It follows that H consists of A , B , and the matching edges $a_i b_i$: any other edge between A and B would be usable. This graph has minimum degree r , again contradicting $(r + 1)$ -connectivity. $\square$

Proof of Theorem 1.1. Let L_1, L_2 be distinct K_r subgraphs and put $s = |V(L_1) \cap V(L_2)|$. If $s \leq r - 2$, delete the common set Z . The graph $G - Z$ is $(r + 1 - s)$ -connected and contains the two disjoint K_{r-s} subgraphs $L_1 - Z$ and $L_2 - Z$. Lemma 2.1 gives a K_{r-s+2}^- model in which every branch set contains a vertex of $(L_1 \cup L_2) - Z$. The vertices of Z , taken as singleton branch sets, are complete to this model and to one another. They complete a K_{r+2}^- model.

It remains that $s = r - 1$. The union $X = V(L_1) \cup V(L_2)$ induces a graph containing K_{r+1}^- . Since an $(r + 1)$ -connected graph has at least $r + 2$ vertices, $G - X$ is nonempty. Let C be a component of $G - X$. If C missed a vertex of X , then $N_G(C)$ would be a cut of order at most r . Thus C is adjacent to every vertex of X . The vertices of X as singleton branch sets, together with C , form a K_{r+2}^- model. $\square$

We use the case $r = 5$ throughout the paper.

Corollary 2.2. *Every six-connected K_7^- -minor-free graph contains at most one K_5 subgraph.*

3. DEGREE-SEVEN NEIGHBOURHOODS

We use two standard facts about contraction-critical graphs. A graph is *k-contraction-critical* if it has chromatic number k and every proper minor is $(k - 1)$ -colourable. A noncomplete k -contraction-critical graph is seven-connected when $k \geq 7$ [7, Satz 6]; see also [12, Theorem 1.8]. Also, if G is k -contraction-critical and $v \in V(G)$, then

$$(3.1) \quad \alpha(G[N(v)]) \leq d(v) - k + 2.$$

This inequality goes back to Dirac [2]; we use the exact formulation in [12, Lemma 1.6(i)].

We use an immediate specialization of Theorem 7 and property (*) in Kriesell and Mohr [6, Theorem 7]. Let a graph J be properly coloured with five colours, and choose one root from each colour class. Let D be a graph on the five roots with at most six edges. If the ends of every edge $xy \in E(D)$ lie in the same bichromatic component of J on their two colours, then J contains pairwise disjoint connected sets, one containing each root, that are adjacent for every edge of D .

Lemma 3.1. *Let G be a seven-chromatic graph such that every proper minor is six-colourable and $K_7^- \not\leq G$. Let v have degree seven, put $H = G[N(v)]$, and let ab be a nonedge of H . If*

$$R = N(v) \setminus \{a, b\},$$

then $G - v - \{a, b\}$ contains a K_5 minor rooted at R .

Proof. Contract the two edges va and vb to one vertex. A six-colouring of the resulting proper minor induces a six-colouring φ of $G - v$ in which a and b have one common colour, say 0. Every vertex of R avoids colour 0. The five vertices of R use the other five colours distinctly, since a colour absent from $N(v)$ could be assigned to v .

Put $F = \overline{H}$. Inequality (3.1) gives $\alpha(H) \leq 2$, so F is triangle-free. If $xy \in E(F[R])$ and x, y have colours i, j , then x, y lie in the same component of the subgraph of $G - v$ induced by colours i, j . Otherwise, interchanging i and j on the component containing x would remove colour i from $N(v)$, allowing that colour at v .

Delete the colour-0 class from $G - v$. The resulting graph is properly five-coloured, with R a transversal of its colour classes. By Mantel's theorem, the triangle-free graph $F[R]$ has at most six edges. The Kriesell and Mohr result, applied with demand graph $F[R]$, gives five disjoint connected sets rooted at R and adjacent for every nonedge of $H[R]$. Every other pair is joined by its edge in $H[R]$, so the five sets form the required rooted K_5 model. $\square$

Theorem 3.2. *Under the hypotheses of Lemma 3.1, every degree-seven vertex v satisfies*

$$(3.2) \quad G[N(v)] \cong K_4 \dot{\cup} K_3 \quad \text{or} \quad K_1 \vee (K_3 \dot{\cup} K_3).$$

In particular, v belongs to a K_5 subgraph.

Proof. Put $H = G[N(v)]$ and $F = \overline{H}$. By (3.1), F is triangle-free. It is nonempty, since otherwise $G[N[v]]$ would contain K_7^- .

Every nonisolated vertex of F has degree at least three. To see this, choose an edge $ab \in E(F)$ and use the rooted model of Lemma 3.1 on $R = N(v) \setminus \{a, b\}$. If $d_F(a) = 1$, then a is adjacent in H to every root; the rooted model, $\{a\}$, and $\{v\}$ form a K_7 model. If $d_F(a) = 2$, with other neighbour x , then triangle-freeness gives $bx \in E(H)$. Absorb b into the branch set rooted at x . Together with $\{a\}$ and $\{v\}$, the resulting seven branch sets have at most one missing adjacency, again a contradiction.

Every nontrivial component of F therefore has minimum degree at least three and hence at least six vertices. Since $|V(F)| = 7$, there is one nontrivial component and at most one isolated vertex. If the component has six vertices, Mantel's theorem and its equality case give $F \cong K_{3,3} \dot{\cup} K_1$.

Suppose instead that the component has seven vertices. No vertex has degree at least five: its independent neighbourhood would contain a vertex of degree at most two. Thus every degree is three or four. The degree sum is even, so some vertex z has degree four. Its four neighbours are independent. Each is adjacent to both vertices outside $N_F[z]$ to have degree at least three, and those two outside vertices are nonadjacent by triangle-freeness. Hence $F \cong K_{3,4}$. Taking complements gives (3.2). $\square$

For a minor-minimal non-six-colourable K_7^- -minor-free graph, Albar's theorem gives $\chi(G) = 7$, while Mader's theorem gives seven-connectivity. Corollary 2.2, together with Theorem 3.2, gives the sharper conclusion below.

Corollary 3.3. *Let G be as in Theorem 1.3. Then G contains at most one K_5 subgraph and every degree-seven vertex v satisfies*

$$G[N(v)] \cong K_4 \dot{\cup} K_3.$$

Proof. The only point not already stated is the exclusion of the second graph in (3.2). In that graph, v belongs to two distinct K_5 subgraphs, which contradicts Corollary 2.2. $\square$

4. ROOTED $K_{4,2}^*$ CLOSURE

For a graph F and $S \subseteq V(F)$, say that (F, S) is *internally k -connected* if there is no separation (A, B) of F with $S \subseteq A$, $B \setminus A \neq \emptyset$, and $|A \cap B| < k$. A Z -rooted $K_{4,2}^*$ model consists of four pairwise disjoint connected branch sets, one containing each member of the four-set Z , and two further branch sets U, V . Each of U, V is adjacent to all four rooted branch sets and to the other; no adjacency between distinct rooted branch sets is required.

We use the rooted bound of Norin and Totschnig [10, Lemma 12].

Theorem 4.1 (Norin and Totschnig). *Let F be a graph and let $Z \subseteq V(F)$ have order four. If (F, Z) is internally four-connected and F has no Z -rooted $K_{4,2}^*$ model, then*

$$(4.1) \quad |E(F)| \leq 4|V(F)| - 10.$$

The next elementary augmentation places one additional prescribed vertex in a non-root branch set.

Lemma 4.2. *Let F be a graph and let $S = Z \dot{\cup} \{x\}$, where $|Z| = 4$. Suppose that (F, S) is internally five-connected. If F has a Z -rooted $K_{4,2}^*$ model, then it has one in which x belongs to a non-root branch set.*

Proof. Choose a Z -rooted model that maximizes $|U| + |V|$ over its two non-root branch sets U, V and, subject to that, minimizes $\sum_{i=1}^4 |R_i|$ over its rooted branch sets $R_1, \dots, R_4$. Write z_i for the root in R_i , and put

$$P_i = \{r \in R_i : r \text{ has a neighbour in } U \cup V\}.$$

We claim that $|P_i| = 1$. Both U and V meet R_i , so P_i is nonempty. If it had at least two vertices, there would be distinct $u, w \in R_i$ such that u meets U and w meets V ; otherwise the two contact sets would be the same singleton. Take a minimal tree in $F[R_i]$ containing z_i, u, w . By the minimal choice of the rooted branch sets, this tree spans R_i . One of u, w , say u , is a leaf different from z_i . Moving u from R_i into U preserves connectivity of both altered branch sets. The former tree edge from u to $R_i \setminus \{u\}$ preserves the contact between R_i and U , while w preserves the contact between R_i and V . Every other required adjacency is unchanged, contradicting the maximality of $|U| + |V|$. Hence $|P_i| = 1$.

No component outside the six branch sets can meet $U \cup V$, since it could be absorbed into whichever of U, V it meets. It follows that

$$(4.2) \quad |N_F(U \cup V) \setminus (U \cup V)| \leq 4,$$

with at most one external neighbour in each rooted branch set. If $x \notin U \cup V$, (4.2) would give a separation of (F, S) of order at most four whose second open side is the nonempty set $U \cup V$. This contradicts internal five-connectivity. Thus $x \in U \cup V$. $\square$

Proof of Theorem 1.2. Choose a four-set $Z \subseteq N_G(v)$ inducing K_4 , choose $x \in N_G(v) \setminus Z$, and put $F = G - v$. Deleting one vertex from a six-connected graph leaves a five-connected graph. In particular, (F, Z) is internally four-connected and $(F, Z \cup \{x\})$ is internally five-connected. Moreover,

$$(4.3) \quad \begin{aligned} |E(F)| &= |E(G)| - d \\ &\geq 4n - 13 = 4|V(F)| - 9. \end{aligned}$$

Theorem 4.1 supplies a Z -rooted $K_{4,2}^*$ model in F , and Lemma 4.2 lets us choose it with x in one non-root branch set.

The four rooted branch sets are pairwise adjacent through the edges of $G[Z] = K_4$. Together with the two non-root branch sets they form a K_6

model. The singleton branch set $\{v\}$ is adjacent to all four rooted branch sets and to the one containing x ; it may miss only the other. These seven branch sets form a K_7^- model in G . $\square$

5. CONSEQUENCES FOR A CRITICAL GRAPH

Jakobsen proved that an n -vertex graph with $n \geq 7$ and at least $\frac{9}{2}n - 12$ edges contains a K_7^- minor or is a $(K_{2,2,2,2}, K_6, 4)$ -cockade [5, Theorem 4]; see also [1, Theorem 2].

Lemma 5.1. *If G is as in Theorem 1.3, with $n = |V(G)|$ and $m = |E(G)|$, then*

$$(5.1) \quad 2m \leq 9n - 25.$$

Proof. The graph G is neither base graph of the cockade family, since both K_6 and $K_{2,2,2,2}$ are six-colourable. A nontrivial cockade has a vertex cut of order four, whereas G is seven-connected. Thus G is not an exceptional cockade. Since it has no K_7^- minor, Jakobsen's theorem gives $m < \frac{9}{2}n - 12$. Integrality yields (5.1). $\square$

The first density bound uses only the linked-clique and neighbourhood results above.

Proposition 5.2. *Let G be as in Theorem 1.3. Then*

$$(5.2) \quad n_7 \leq 5, \quad |E(G)| \geq 4|V(G)| - 2.$$

Proof. By Corollary 3.3, every degree-seven vertex lies in the unique possible K_5 , so $n_7 \leq 5$. Seven-connectivity gives minimum degree at least seven, and every vertex not counted by n_7 has degree at least eight. Hence

$$2|E(G)| \geq 7n_7 + 8(|V(G)| - n_7) = 8|V(G)| - n_7 \geq 8|V(G)| - 5.$$

The left side is even, so it is at least $8|V(G)| - 4$, proving (5.2). $\square$

Proof of Theorem 1.3. Albar's theorem gives $\chi(G) = 7$, and every proper minor is six-colourable by the choice of G . The graph is noncomplete because it has no K_7^- minor, so Mader's theorem gives seven-connectivity.

Suppose that v has degree seven. Corollary 3.3 shows that $G[N(v)]$ contains a K_4 . Proposition 5.2 gives

$$|E(G)| \geq 4|V(G)| - 2 > 4|V(G)| - 6,$$

which is more than the threshold in Theorem 1.2 when $d_G(v) = 7$. That theorem gives the forbidden K_7^- minor. Thus $n_7 = 0$. Seven-connectivity now gives $\delta(G) \geq 8$, and degree summation gives $|E(G)| \geq 4|V(G)|$.

Put $q = |E(G)| - 4|V(G)|$, so $q \geq 0$. Suppose that a K_5 subgraph has vertex set K . For each $w \in K$, the neighbourhood of w contains the K_4 on $K \setminus \{w\}$. Since G has no K_7^- minor, the contrapositive of Theorem 1.2 gives

$$4|V(G)| + q < 4|V(G)| + d_G(w) - 13,$$

and hence $d_G(w) \geq q + 14$. The five vertices of K would contribute at least $5(q + 6)$ to

$$\sum_{z \in V(G)} (d_G(z) - 8) = 2q,$$

which is impossible. Thus G contains no K_5 subgraph.

Finally, Lemma 5.1 and the exact degree identity give

$$25 \leq 9|V(G)| - 2|E(G)| = n_8 - \tau.$$

Therefore $n_8 \geq 25 + \tau$. If a degree-eight vertex had a K_4 in its neighbourhood, it would belong to a K_5 , which has just been excluded. $\square$

6. FOUR INDEPENDENT DEGREE-EIGHT VERTICES

We prove Theorem 1.4 using $R(5, 4) = 25$, which follows by complement symmetry from the theorem of McKay and Radziszowski [9], and the rooted K_4 theorem of Fabila-Monroy and Wood [3].

Lemma 6.1. *Let G be as in Theorem 1.3. Then G contains an independent four-set U of degree-eight vertices. For every such set, $H = G - U$ is three-connected, nonplanar, and six-chromatic. Moreover $\kappa(G) \in \{7, 8\}$, and if $\kappa(G) = 8$ then H is four-connected.*

Proof. By Theorem 1.3, at least twenty-five vertices have degree eight and G contains no K_5 subgraph. The Ramsey equality $R(5, 4) = 25$ therefore supplies the required independent set U .

Deleting four vertices lowers connectivity by at most four, so H is three-connected, and is four-connected when G is eight-connected. For each $u \in U$, the set $N_G(u)$ separates u from the other members of U , whence $\kappa(G) \leq 8$.

The independence and degrees of the four vertices give

$$|E(H)| = |E(G)| - 32 \geq 4|V(H)| - 16.$$

Since $|V(H)| \geq 21$, this exceeds $3|V(H)| - 6$, so H is nonplanar. It is six-colourable as a proper minor of G . A five-colouring of H would extend to a six-colouring of G by assigning one new colour to all of U . Thus $\chi(H) = 6$. $\square$

For the rest of this section, let G be as in Theorem 1.3, let U be an independent four-set of degree-eight vertices, and put $H = G - U$. Statements involving $r, \varphi, x_1, x_2, x_3, x_4$ use the choices in Theorem 6.2; in its web outcome, T, C , and $S = U \dot{\cup} T$ have the meanings fixed there.

6.1. The rooted-web alternative. A web rooted at four nominated vertices has a plane skeleton with the roots on its outer face; additional clique sets may be placed behind facial triangles and joined to all three vertices of the corresponding triangle. We call the three vertices of such a triangle a *facial three-set*; they need not induce a triangle in a spanning subgraph of the web. Fabila-Monroy and Wood proved that a three-connected graph has a K_4 minor rooted at four nominated vertices if and only if it is not a

spanning subgraph of a web rooted at those vertices [3, Theorem 8]. Edges of a facial triangle in the containing web need not be edges of the spanning subgraph.

Theorem 6.2. *Fix $r \in U$ and a proper six-colouring φ of $G - r$. Choose four colours occurring exactly once on $N_G(r)$, and let x_1, x_2, x_3, x_4 be their unique representatives. Exactly one of the following holds.*

- (1) *H contains a K_4 model rooted at x_1, x_2, x_3, x_4 . Together with $\{r\}$, this is a K_5 model in $G - (U - \{r\})$.*
- (2) *H is a spanning subgraph of a web rooted at x_1, x_2, x_3, x_4 . Some facial three-set T and some nonempty connected set $C \subseteq V(H) - T$ satisfy*

$$(6.1) \quad N_H(C) = T, \quad N_G(C) = S := U \dot{\cup} T.$$

No x_i belongs to C , and at least one x_i belongs to a component of $G - S$ other than C . Every component D of $G - S$ satisfies $N_G(D) = S$.

Proof. Every colour occurs on $N_G(r)$, since otherwise φ extends to r . Six positive multiplicities summing to eight include at least four ones. Since U is independent, the corresponding representatives lie in H .

Apply the rooted theorem quoted above to H . Its rooted-model outcome gives item 1 because r is adjacent to the nominated vertex in every rooted branch set. All five branch sets avoid $U - \{r\}$.

Otherwise H is a spanning subgraph of the stated web. Since H is non-planar, some clique set X_T behind a facial three-set T is nonempty in H . Let C be a component of $H[X_T]$. Web containment gives $N_H(C) \subseteq T$. At least one outer root lies outside T , so three-connectivity forces $N_H(C) = T$.

In G , every neighbour of C lies in $T \cup U$. The outer root chosen outside T also lies outside $C \cup T \cup U$, so $N_G(C)$ separates it from C . Seven-connectivity therefore forces $N_G(C) = T \cup U$, proving (6.1). The vertices added behind T are disjoint from the skeleton, so no x_i belongs to C , and one lies outside the three-set T . Finally, any component of $G - S$ whose neighbourhood were a proper subset of S would contradict seven-connectivity. Thus every component D of $G - S$ has $N_G(D) = S$. $\square$

Corollary 6.3. *If $\kappa(G) = 8$, item 1 of Theorem 6.2 holds for every choice of r , φ , and the four singleton-colour neighbours.*

Proof. Item 2 gives a seven-vertex separator. $\square$

6.2. Minimum lift order. For $u \in U$, write $N_H(u) = N_G(u) \cap V(H)$. For a separation $p = (A, B)$ of H , write

$$L_p = A - B, \quad R_p = B - A, \quad S_p = A \cap B,$$

and say that u crosses p if it has a neighbour in both L_p and R_p . Define

$$(6.2) \quad \begin{aligned} C_U(p) &= \{u \in U : N_H(u) \cap L_p \neq \emptyset \neq N_H(u) \cap R_p\}, \\ \lambda_U(p) &= |S_p| + |C_U(p)|. \end{aligned}$$

A lift of p is *trace-preserving* if its intersections with H are exactly A and B .

Theorem 6.4. *For every separation p of H , the number $\lambda_U(p)$ is the minimum order of a trace-preserving lift to G . The function λ_U is symmetric and submodular. More precisely, for separations $p = (A, B)$ and $q = (C, D)$, put*

$$p \wedge q = (A \cap C, B \cup D), \quad p \vee q = (A \cup C, B \cap D).$$

Then

$$(6.3) \quad \lambda_U(p \wedge q) + \lambda_U(p \vee q) \leq \lambda_U(p) + \lambda_U(q).$$

If equality holds, equality holds separately for the crossing indicator of every $u \in U$. If G is k -connected and p is proper, then $\lambda_U(p) \geq k$.

Proof. Every lift must put S_p and every member of $C_U(p)$ in its separator. Conversely, put precisely those vertices there; put each remaining member of U on a side containing all its neighbours outside S_p . The independence of U creates no crossing edge. This proves the minimum formula and also symmetry.

The ordinary separator terms satisfy the modular identity

$$(6.4) \quad |S_p| + |S_q| = |S_{p \wedge q}| + |S_{p \vee q}|.$$

For one fixed u , its crossing indicator is submodular. Indeed,

$$\begin{aligned} L_{p \wedge q} &= L_p \cap L_q, & R_{p \wedge q} &= R_p \cup R_q, \\ L_{p \vee q} &= L_p \cup L_q, & R_{p \vee q} &= R_p \cap R_q. \end{aligned}$$

If u crosses both corners it crosses both inputs; if it crosses only one corner it crosses at least one input. Summing over U and using (6.4) proves (6.3), and equality in the sum implies equality for each $u \in U$. A proper separation of H has a proper minimum lift, so k -connectivity gives the last assertion. $\square$

For a family of oriented separations, fix vertices a, b such that $a \in L_p$ and $b \in R_p$ for every separation p in the family.

Corollary 6.5. *Assume $\kappa(G) = 7$. Every proper three-separation p of H is crossed by all four members of U and lifts to an order-seven separation with separator $U \dot{\cup} S_p$, where S_p is its three-vertex separator in H . Conversely, the trace of every order-seven separation of this form is a proper three-separation of H crossed by all four members of U .*

For fixed a, b as above, these oriented separations form a finite meet/join lattice. Its inclusion-minimum open side containing a is connected.

Proof. For a proper three-separation, Theorem 6.4 gives $3 + |C_U(p)| \geq 7$, so all four members of U cross and equality holds. Conversely, every vertex in the separator of an order-seven separation of a seven-connected graph has a neighbour in both open sides; otherwise the other six separator vertices would still separate one side.

If two order-seven separations have the same fixed vertices a, b , their meet and join are proper. Connectivity bounds each corner order below by seven, while submodularity bounds their sum above by fourteen. Both corners therefore have order seven. Equality in (6.3), together with the pointwise equality clause of Theorem 6.4, makes every member of U cross both corners. Hence each corner has H -separator order $7 - 4 = 3$, so repeated meets stay in the stated family and give the minimum open side. If that side were disconnected, the component containing a would have neighbourhood contained in the same seven-set. Seven-connectivity makes that neighbourhood the whole separator, producing a smaller open side containing a , a contradiction. $\square$

Corollary 6.6. *Let $p_1, \dots, p_m$ be proper three-separations of H whose trace-preserving lifts have order seven as in Corollary 6.5. For $\sigma \in \{+, -\}^m$, let R_σ be the intersection of the open sides selected by its signs. If both R_σ and $R_{-\sigma}$ are nonempty, then every $u \in U$ has a neighbour in R_σ .*

Consequently at most eight sign regions whose antipodal regions are also nonempty can occur. If all four regions of two separations are nonempty, every member of U meets all four. If all eight regions of three separations are nonempty, every member of U has exactly one neighbour in each region and no separator neighbour. Four separations cannot have all sixteen sign regions nonempty.

Proof. Choose $a \in R_\sigma$ and $b \in R_{-\sigma}$, orient the separations accordingly, and take repeated meets. Corollary 6.5 gives another order-seven separation whose selected open side is R_σ ; every member of U crosses it. Distinct regions are disjoint and every member of U has degree eight, giving all the conclusions. $\square$

6.3. The colour-labelled web obstruction.

Theorem 6.7. *Assume the web outcome and list x_1, x_2, x_3, x_4 in their cyclic order on the outer face. Put $c_i = \varphi(x_i)$. Every pair x_i, x_j lies in the same (c_i, c_j) -component of $G - r$. Let K_{13} be the (c_1, c_3) -component containing x_1, x_3 , and define K_{24} analogously. Then K_{13} and K_{24} are vertex-disjoint, and at least one contains a member of $U - \{r\}$.*

If Δ is the set of two unselected colours, some $s \in U - \{r\}$ sees both colours of Δ in H .

Proof. If x_i, x_j were in different (c_i, c_j) -components, interchange the two colours on the component containing x_i . Since x_i and x_j are the unique neighbours of r in their respective colours, colour c_i would then be absent from $N_G(r)$, and could be assigned to r . This is impossible.

The two components K_{13} and K_{24} use disjoint colour pairs and hence are vertex-disjoint. If both avoided $U - \{r\}$, they would give in H two disjoint paths joining alternating outer pairs, contrary to the Two Paths description of a web [3, Lemma 2 and the first proof of Lemma 4].

Suppose finally that every $s \in U - \{r\}$ missed in $N_H(s)$ some colour $\delta_s \in \Delta$. Recolour each s with δ_s . These changes are simultaneous and proper because U is independent, and they do not change the colours on $N_G(r)$. Repeating the preceding argument, both alternating components now avoid $U - \{r\}$, the same contradiction. $\square$

6.4. The facial three-set.

Corollary 6.8. *In the web outcome, $G[T]$ is not a triangle.*

Proof. Suppose that $G[T]$ is a triangle, say $T = \{t_1, t_2, t_3\}$. Let $R = V(G) - (C \cup S)$. Since G is seven-connected, every component of $G - S$ has neighbourhood S , so both $C \cup U$ and $R \cup U$ are connected. Contract a spanning tree of $C \cup U$, six-colour the proper minor, and pull the colouring back only to $G[R \cup S]$. Its colour classes on S are

$$(6.5) \quad U \mid \{t_1\} \mid \{t_2\} \mid \{t_3\}.$$

Contracting $R \cup U$ instead produces the same colour classes on $G[C \cup S]$. A permutation of colour names aligns the two colourings, which then glue to a six-colouring of G , a contradiction. $\square$

6.5. A colouring response to an edge deletion at the web separator.

Theorem 6.9 (Controlled paths or a smaller component). *Assume outcome 2 of Theorem 6.2. Choose $x = x_j$ in a component D of $G - S$ distinct from C , put $\gamma = \varphi(x)$, and let $e = rx$. Thus $S = U \dot{\cup} T$ and $N_G(C) = N_G(D) = S$. Extend φ to r by assigning colour γ to r , and call the resulting map d . Then d is a proper six-colouring of $G - e$.*

For every colour $\beta \neq \gamma$, the vertices x and r lie in the same (γ, β) -component of $G - e$. Choose a simple path from x to r in this component, orient it from x , and stop it at its first vertex $q_\beta \in S$. Write xy_β for its first edge; the stopped path has all internal vertices in D . Put

$$Q_0 = \{r\} \cup \{q_\beta : \beta \neq \gamma\}.$$

The five vertices y_β are distinct. At least one of the following holds.

- (1) *There is a set $Q \subseteq S$ with $Q_0 \subseteq Q$ and $|Q| \leq 6$, and five paths from x to Q which begin with the prescribed edges xy_β and are pairwise vertex-disjoint outside $\{x\} \cup Q$. Their ends in Q need not be distinct.*
- (2) *One has $|Q_0| \leq 5$, and there are a five-set $Q \subseteq S$ containing Q_0 , a four-set $Z \subseteq D - \{x\}$, and a nonempty connected set $A \subseteq D - (\{x\} \cup Z)$ such that*

$$(6.6) \quad N_G(A) = (S - Q) \dot{\cup} \{x\} \dot{\cup} Z.$$

In particular, $N_G(A)$ is a seven-vertex separator and $|A| < |D|$. The restriction of d to $G[A \cup N_G(A)]$ is a proper six-colouring. If

$$J = \{z \in N_G(A) : d(z) = \gamma\},$$

then $|J| \geq 2$. In this colouring, and also in some proper six-colouring of $G - A$, the set J is exactly one colour class on $N_G(A)$.

Independently of which alternative occurs, there is a fan from x to six distinct vertices of S , consisting of the edge xr and five paths beginning with the prescribed edges xy_β . Delete x from the five non-direct paths and denote the resulting disjoint connected sets by L_β . The contact graph of these five sets, in which two are adjacent when they have an edge between them in G , has at most eight edges.

Proof. The vertex x is the unique γ -coloured neighbour of r . Consequently rx is the only monochromatic edge after restoring r , so d is proper on $G - e$.

Fix $\beta \neq \gamma$. If x and r belonged to different (γ, β) -components, interchange the two colours on the component containing x . Every β -coloured neighbour of r lies in the component containing r , and x is its only γ -coloured neighbour. The interchange would therefore repair rx without creating another conflict at r , giving a six-colouring of G . This contradiction proves the claimed connection. Since $d(x) = \gamma$, each y_β has colour β ; hence the five vertices y_β , and the five prescribed edges, are distinct.

Let $I = \{\beta : y_\beta = q_\beta\}$ be the colours whose prescribed path is already a boundary edge, and put $q = |I|$.

Suppose first that $|Q_0| \leq 5$, and enlarge Q_0 to a five-set $Q \subseteq S$. Retain the q direct paths. In

$$G[(D - \{x\}) \cup Q]$$

seek paths from the other $5 - q$ vertices y_β to Q , using every source, with unit vertex capacity in $D - \{x\}$ and unrestricted capacity at Q . If such a routing exists, prepending the prescribed first edges gives outcome 1.

Otherwise, the capacitated form of Menger's theorem gives a set $Z \subseteq D - \{x\}$ with $|Z| \leq 4 - q$ meeting every such path. Some remaining source avoids Z ; let A be its component in $G[D - (\{x\} \cup Z)]$. It has no neighbour in Q , and therefore

$$N_G(A) \subseteq (S - Q) \cup \{x\} \cup Z.$$

The right side has order at most $7 - q$. The nonempty component C lies on the other side, so this is a genuine separation. Seven-connectivity forces $q = 0$, $|Z| = 4$, and equality throughout, proving (6.6).

For each $\beta \neq \gamma$, the original stopped path with x deleted joins y_β to Q , and hence meets Z . Choose one such intersection vertex z_β . Five choices in the four-set Z give $z_\beta = z_{\beta'}$ for two distinct colours β, β' . A common vertex of a γ, β path and a γ, β' path has colour γ . Thus J contains both this vertex and x , so $|J| \geq 2$.

The only edge on which d is improper in G is rx . Since $r \in Q$, this edge is absent from $G[A \cup N_G(A)]$, so the restriction of d to that graph is proper.

The set $A \cup J$ is connected, and J is independent. Contract a spanning tree of $G[A \cup J]$ and six-colour the resulting proper minor. Pull the colouring back to $G - A$, assigning every member of J the colour of the contraction image. That image is adjacent to every vertex of $N_G(A) - J$, so J is exactly one colour class on $N_G(A)$.

It remains to treat $|Q_0| = 6$. Set $Q = Q_0$ and retain the q direct paths as above. Failure to route the remaining sources to Q would give a set Z of order at most $4 - q$ and a nonempty connected set $A \subseteq D - \{x\}$ satisfying

$$N_G(A) \subseteq (S - Q) \cup \{x\} \cup Z.$$

This set has order at most $1 + 1 + (4 - q) \leq 6$, again separates A from C , and contradicts seven-connectivity. Outcome 1 therefore holds.

For the distinct-ended fan, again retain the q prescribed edges which already end in S . Put

$$W = S - (\{r\} \cup \{y_\beta : \beta \in I\}).$$

Working in $G[(D - \{x\}) \cup W]$, route the remaining $5 - q$ vertices y_β to distinct members of W . If this routing failed, Menger's theorem would give a separator of order at most $4 - q$. Since there are $5 - q$ sources and $6 - q$ targets, at least one of each survives. A surviving source component would then have its neighbourhood contained in that separator together with x , r , and the q retained boundary ends, a set of order at most six. A surviving target lies on the other side, contradicting seven-connectivity. Prepending the prescribed first edges and adjoining xr gives the fan.

Finally suppose that the contact graph of the five sets L_β had at least nine edges. The seven disjoint connected branch sets

$$\{x\}, \quad C \cup \{r\}, \quad L_\beta \quad (\beta \neq \gamma)$$

would form a K_7^- model. The first set meets every L_β through its prescribed first edge. The set $C \cup \{r\}$ is connected, meets $\{x\}$ through rx , and meets every L_β through its distinct end in $S - \{r\}$ because $N_G(C) = S$. Among the five remaining branch sets at most one adjacency is missing. This contradicts the choice of G , and proves the contact bound. $\square$

Corollary 6.10 (Restoring one vertex of U). *In the web outcome, for every $s \in U - \{r\}$ the graph $G - (U - \{s\})$ contains a K_4 model rooted at x_1, x_2, x_3, x_4 , and every such model uses s .*

Proof. Deleting the three vertices of $U - \{s\}$ from the seven-connected graph G leaves the four-connected graph $G - (U - \{s\})$. It is nonplanar because it contains H . The four-connected rooted- K_4 theorem of Fabila-Monroy and Wood [3, Theorem 6] therefore supplies the rooted model. The graph H has no such model in the web outcome, so every model in $G - (U - \{s\})$ must use s . $\square$

The web case remains open: the smaller separator in Theorem 6.9 need not contain U , and the fan paths are not assigned to prescribed branch sets.

One must retain U when passing to the smaller component or absorb the paths into one rooted model.

7. EXTREMAL REDUCTION AND SCOPE

Corollary 7.1 (Extremal reduction). *Suppose that every seven-connected graph H satisfying*

$$|E(H)| \geq 4|V(H)|$$

contains a K_7^- minor. Then every K_7^- -minor-free graph is six-colourable.

Proof. Otherwise, choose a minor-minimal non-six-colourable K_7^- -minor-free graph G . By Theorem 1.3, the graph G is seven-connected and $|E(G)| \geq 4|V(G)|$. The assumed extremal statement therefore gives $K_7^- \preceq G$, a contradiction. $\square$

The extremal hypothesis remains open, so this paper proves neither the Norin and Totschnig conjecture nor Hadwiger’s conjecture for $t = 7$.

STATEMENT ON AI USE AND INDEPENDENCE

GPT-5.6 Sol, an OpenAI model, contributed to theorem discovery, proofs, verification, and drafting. Cavit Erginsoy directed the work and made the final decisions. Erginsoy is not affiliated with OpenAI, and OpenAI neither sponsored nor verified the work.

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